

Lecture 4 — Bases, Orthogonality, and Projection

18.085 Computational Science and Engineering I • MW 9:30–11, 6-120 • Monday, September 21, 2026

After this lecture you should be able to

- Read $\mathbf{Ax} = \mathbf{b}$ as the search for coefficients that build $\mathbf{b}$ out of the columns of $\mathbf{A}$.
- Recognise when a set of vectors forms a basis, and write one vector in three different bases.
- Test orthogonality with the inner product and connect it to the angle between two vectors.
- Compute coefficients in an orthogonal basis by projection instead of by elimination.

The same equation, asked differently

For two lectures we have treated $\mathbf{Ax} = \mathbf{b}$ as a computation. Today we treat it as a *question about geometry*, and the answer will change what we do with it. The mathematics is not new — we solve a 2×2 system on the first page — but the reading is, and it is the reading that carries us into least squares, Fourier series, and the SVD later in the term.

1 A 2×2 system, read geometrically

Consider the linear system

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

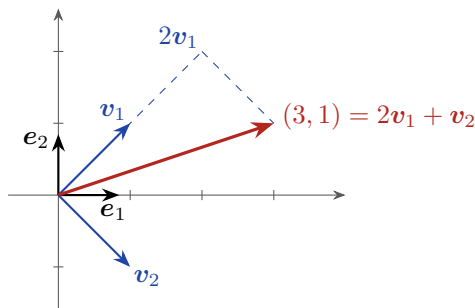
We know how to solve this, of course. I want to draw attention to its geometric interpretation instead.

Recall from Lecture 1 that a matrix–vector product can be understood as a linear combination of the *columns* of the matrix:

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = a \underbrace{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}_{\mathbf{v}_1} + b \underbrace{\begin{bmatrix} 1 \\ -1 \end{bmatrix}}_{\mathbf{v}_2} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

So the system is not really asking for two numbers a and b . It is asking: *how much of $\mathbf{v}_1$ and how much of $\mathbf{v}_2$ do I need in order to land on $(3, 1)$?*

The solution is $a = 2$, $b = 1$.



Geometrically, we have found the weights, or coefficients, needed to represent the vector $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ with the vectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

2 Bases

Nothing was special about $(3, 1)$. We can see that *any* point in the two-dimensional plane can be represented by a **unique** linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$. That is the definition we want.

Definition 2.1. A set of vectors is a **basis** for a linear space if every vector in the space is a linear combination of them, and that combination is unique. Here $\mathbf{v}_1$ and $\mathbf{v}_2$ form a basis for the two-dimensional linear space of the plane.

In fact, when we write down the vectors $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$ or $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ or $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$, we are already implicitly using a specific basis — the **Cartesian basis** — to represent points on the plane. The Cartesian basis vectors point along the axes as drawn:

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Coordinates are a report, not the thing

A vector is a geometric object — an arrow in the plane. The column of numbers you write down is a *report* of that arrow relative to a basis you have chosen, usually without noticing. Change the basis and the numbers change; the arrow does not. Most of the clever methods in this course amount to choosing a basis in which the numbers come out simple.

3 A third basis, and the general statement

Let us take a basis with no symmetry at all:

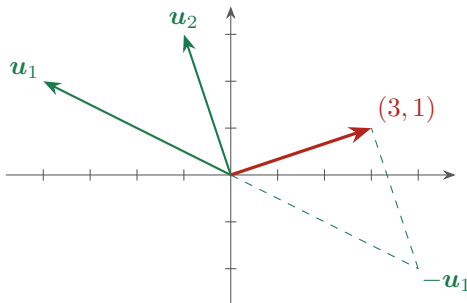
$$\mathbf{u}_1 = \begin{bmatrix} -4 \\ 2 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}.$$

To find out how to represent the vector $(3, 1)$ using $\mathbf{u}_1$ and $\mathbf{u}_2$, we just need to solve the linear system whose *columns* are those two vectors:

$$\begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} -4 & -1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \quad \implies \quad c = -1, \quad d = 1,$$

that is,

$$-\mathbf{u}_1 + \mathbf{u}_2 = \begin{bmatrix} 4 \\ -2 \end{bmatrix} + \begin{bmatrix} -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$



The two constructions in the two pictures land on the same red arrow. Only the scaffolding differs. That is the general moral:

What a linear system is really asking

Another interpretation of solving a linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$: find coefficients $x_1, \dots, x_m$ for the columns $\mathbf{a}_1, \dots, \mathbf{a}_m$ of $\mathbf{A}$ such that the linear combination

$$x_1\mathbf{a}_1 + \dots + x_m\mathbf{a}_m = \mathbf{b}.$$

Solving is a change of coordinates: you are re-expressing $\mathbf{b}$ in the basis given by the columns of $\mathbf{A}$.

4 Orthogonality

Let us compare the three sets of bases we now have:

$$\underbrace{\{\mathbf{e}_1, \mathbf{e}_2\}}_{\text{orthonormal}}, \quad \{\mathbf{v}_1, \mathbf{v}_2\}, \quad \{\mathbf{u}_1, \mathbf{u}_2\}.$$

The first two are **orthogonal**; the first is orthogonal *and* normalised. The key difference between $\{\mathbf{u}_1, \mathbf{u}_2\}$ and the other two is exactly that:

$$\mathbf{e}_1 \perp \mathbf{e}_2, \quad \mathbf{v}_1 \perp \mathbf{v}_2, \quad \text{but NOT for } \mathbf{u}_1, \mathbf{u}_2.$$

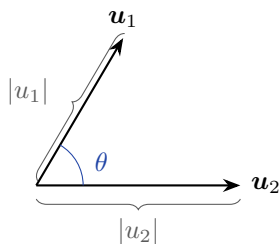
From a linear algebra perspective, orthogonality is reflected by the **inner product**:

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}^T \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 1 \cdot 1 + 1 \cdot (-1) = 0, \quad \text{whereas} \quad \mathbf{u}_1 \cdot \mathbf{u}_2 = (-4)(-1) + (2)(3) = 10 \neq 0.$$

Geometrically, for any two vectors,

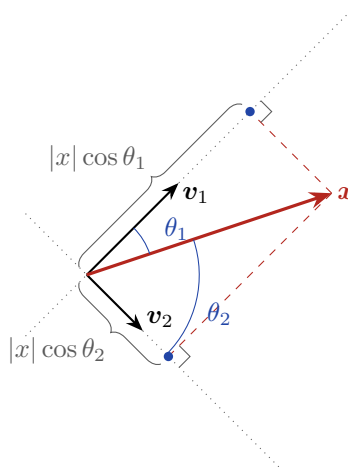
$$\mathbf{u}_1 \cdot \mathbf{u}_2 = \mathbf{u}_1^T \mathbf{u}_2 = |\mathbf{u}_1| |\mathbf{u}_2| \cos \theta.$$

Given $|\mathbf{u}_1|, |\mathbf{u}_2| > 0$, the statement $\mathbf{u}_1 \cdot \mathbf{u}_2 = 0$ therefore implies $\cos \theta = 0$, so $\theta = \pm \frac{\pi}{2}$ — orthogonal, in the ordinary sense of the word.



5 Projection: coefficients without elimination

Here is why we care. The nice thing about orthogonal bases is that finding the coefficients of a vector in such a basis can be done by **projection**, instead of by solving a linear system directly.



Suppose $\mathbf{v}_1$ and $\mathbf{v}_2$ are orthogonal. Drop a perpendicular from $\mathbf{x}$ onto each axis. The component along $\mathbf{v}_1$ has length $|x| \cos \theta_1$, so it contributes $|x| \cos \theta_1$ divided by the length of $\mathbf{v}_1$ copies of $\mathbf{v}_1$, and likewise for $\mathbf{v}_2$:

$$\mathbf{x} = \frac{|x| \cos \theta_1}{|v_1|} \mathbf{v}_1 + \frac{|x| \cos \theta_2}{|v_2|} \mathbf{v}_2.$$

Multiply each numerator and denominator by the length of the corresponding basis vector,

$$\mathbf{x} = \frac{|v_1| |x| \cos \theta_1}{|v_1|^2} \mathbf{v}_1 + \frac{|v_2| |x| \cos \theta_2}{|v_2|^2} \mathbf{v}_2,$$

and recognise every piece as an inner product:

$$\mathbf{x} = \underbrace{\left(\frac{\mathbf{v}_1^\top \mathbf{x}}{\mathbf{v}_1^\top \mathbf{v}_1} \right)}_{\text{coefficient of } \mathbf{v}_1} \mathbf{v}_1 + \underbrace{\left(\frac{\mathbf{v}_2^\top \mathbf{x}}{\mathbf{v}_2^\top \mathbf{v}_2} \right)}_{\text{coefficient of } \mathbf{v}_2} \mathbf{v}_2.$$

Test it on the example we started with. With $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$, we found $a\mathbf{v}_1 + b\mathbf{v}_2 = \mathbf{x}$ with $a = 2$, $b = 1$. And indeed

$$a = \frac{\mathbf{v}_1^\top \mathbf{x}}{\mathbf{v}_1^\top \mathbf{v}_1} = \frac{4}{2} = 2, \quad b = \frac{\mathbf{v}_2^\top \mathbf{x}}{\mathbf{v}_2^\top \mathbf{v}_2} = \frac{2}{2} = 1.$$

No elimination, no back substitution — two dot products. If the basis is also *normalised*, so that the basis vectors have length 1, the denominators are 1 and we can leave them out entirely.

Orthogonality decouples

In a general basis the coefficients are entangled: you cannot know a without also solving for b , which is why elimination is needed. In an orthogonal basis each coefficient is computed by one formula that never mentions the other basis vectors. Every appearance of orthogonality later in this course — QR, least squares, Fourier series, the SVD — is bought for that reason.

The projection formula is not a general formula

It computes coefficients *in an orthogonal basis*, and the derivation used orthogonality in the step where the perpendicular from $\mathbf{x}$ onto the $\mathbf{v}_1$ axis was allowed to ignore $\mathbf{v}_2$ entirely. Run

it blindly on the non-orthogonal pair $\mathbf{u}_1, \mathbf{u}_2$ and you get $\mathbf{u}_1^\top \mathbf{x} / \mathbf{u}_1^\top \mathbf{u}_1 = -10/20 = -\frac{1}{2}$, which is not the coefficient $c = -1$ we computed above. For a non-orthogonal basis you must solve the system.

6 The Cartesian basis, and why the identity matrix looks like that

Apply the formula to the basis we have been using all along. The Cartesian basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ is orthogonal *and* normalised, so the denominators disappear and

$$\mathbf{x} = \underbrace{(\mathbf{e}_1^\top \mathbf{x})}_{\substack{\text{first entry} \\ \text{of } \mathbf{x}, x(1)}} \mathbf{e}_1 + \underbrace{(\mathbf{e}_2^\top \mathbf{x})}_{x(2)} \mathbf{e}_2 = \begin{bmatrix} | & | \\ \mathbf{e}_1 & \mathbf{e}_2 \\ | & | \end{bmatrix} \begin{bmatrix} x(1) \\ x(2) \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{\text{the identity } \mathbf{I}} \mathbf{x}.$$

So the projection formula, applied to the standard basis, says $\mathbf{x} = \mathbf{I}\mathbf{x}$ — which is true, and is meant to be reassuring rather than surprising. It says that “reading off the entries of a vector” is not a primitive operation at all: it is projection onto a particular orthonormal basis, one we chose so long ago that we stopped noticing we had chosen it.

Where the four fundamental subspaces come in

Two subspaces have already appeared without being named. The set of vectors reachable as $\mathbf{A}\mathbf{x}$ — everything you can build from the columns — is the **column space**, and it is what decides whether $\mathbf{A}\mathbf{x} = \mathbf{b}$ has a solution at all. The set of $\mathbf{x}$ with $\mathbf{A}\mathbf{x} = \mathbf{0}$ is the **nullspace**, and we met it in Lecture 2 as the rigid translation $(1, 1, 1, 1)$ of a free-free rod. Doing the same two constructions for $\mathbf{A}^\top$ gives two more, and those four subspaces — column space and nullspace of $\mathbf{A}$, column space and nullspace of $\mathbf{A}^\top$ — are the standard bookkeeping for every solvability question we will meet.

Where this is going

Next time we go back to $-u'' = f$ and ask what happens when the load is concentrated at a single point: the delta function, and the Green’s function that is the response to it. That is this lecture’s idea in continuous costume — decompose the thing you want into simple pieces, solve for each piece, and add the results up.

Reading

Strang, *Computational Science and Engineering*, § (TBD). Bases and orthogonality are the linear-algebra background assumed throughout Chapter 1; the projection formula returns in earnest with least squares and QR later in the term.